AN AI-POWERED RESUME ANALYZER FOR INTELLIGENT CAREER GUIDANCE ON CLOUD-NATIVE MICROSERVICE ARCHITECTURE

CLOUD COMPUTING

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BONAFIDE CERTIFICATE

Certified that this project report titled "AN AI-POWERED RESUME ANALYZER FOR INTELLIGENT CAREER GUIDANCE" is the bonafide work of SHRADHA S (2307010311), ROSHINI V (230701270), DHEERAJ NELLORE (230701077) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The job market is increasingly competitive, with recruiters spending an average of six seconds reviewing a resume before making a decision. Studies indicate that over 75% of resumes are rejected by Applicant Tracking Systems (ATS) before a human ever reads them, primarily due to formatting issues, missing keywords, and poor structural alignment with job descriptions. Students and fresh graduates are disproportionately affected, lacking professional guidance to craft resumes that clear automated filters and impress hiring managers. Existing tools such as generic grammar checkers or basic resume templates do not offer intelligent, context-aware feedback tailored to specific job roles or industries. There is a clear need for a dedicated AI-powered platform that bridges this gap — one that evaluates a resume holistically, compares it against job descriptions, and provides actionable, intelligent improvement suggestions. This project delivers a secure, cloud-native AI Resume Analyzer platform that leverages Google Gemini's generative AI capabilities to perform deep resume analysis. Users upload their resume in PDF or DOCX format along with a target job description. The system extracts and parses the resume content, sends it to the Gemini API with a structured prompt, and receives detailed, role-specific feedback covering skills alignment, keyword optimization, formatting quality, and content gaps. The platform is built using a modern microservice architecture. The frontend is developed with HTML, CSS, and JavaScript and hosted on Azure Static Web Apps for global CDN delivery. The Node.js backend is deployed on Azure App Services, handling all business logic including file processing, AI orchestration, and user session management. Azure Cosmos DB serves as the primary NoSQL database for storing user profiles, analysis history, and job description data. Containerization is achieved via Docker and orchestrated using Azure Kubernetes Service (AKS), with container images stored in Azure Container Registry (ACR). Infrastructure is fully automated using Terraform as Infrastructure-as-Code (IaC), enabling reproducible and environment-consistent provisioning of all Azure resources. A complete CI/CD pipeline implemented with GitHub Actions automates build, test, and deployment workflows. Azure Application Insights provides real-time monitoring and observability, while Azure Key Vault secures all sensitive credentials and API keys. The expected outcome is a responsive, scalable, and cost-effective resume intelligence platform that helps students and job seekers improve their application quality, increase ATS pass rates by an estimated 50-60%, and gain confidence in their career documents. By combining AI intelligence with cloud-native deployment best practices, the platform promotes equitable access to professional career guidance and positions students for success in an increasingly automated hiring landscape.

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CHAPTER 1 - INTRODUCTION

1.1 PROBLEM STATEMENT

In today's hyper-competitive job market, a well-crafted resume is the single most critical document in a candidate's career journey. Recruiters and hiring managers are overwhelmed with applications, spending an average of just six seconds on each resume before deciding whether to proceed. Compounding this challenge, most large organizations now deploy Applicant Tracking Systems (ATS) that automatically screen resumes before any human review, filtering candidates based on keyword matching, formatting compliance, and structural alignment with job descriptions.

Key challenges that motivated this project include:

- High ATS Rejection Rates: Over 75% of resumes submitted to large organizations are automatically rejected by ATS software before reaching a human recruiter. Reasons include improper formatting, missing industry-specific keywords, and failure to align content with the target job description. This creates an invisible barrier that disproportionately disadvantages students and entry-level candidates who lack insider knowledge of these systems.
- Lack of Personalized Feedback: Generic resume templates and basic grammar-check tools do not offer role-specific, context-aware analysis. A resume suitable for a software engineering role differs fundamentally from one targeting a data science or product management position, yet most available tools treat all resumes identically.
- Cost and Accessibility Barriers: Professional resume review services charge anywhere from ₹2,000 to ₹10,000 per session, placing them out of reach for most students and early-career professionals. Free online tools offer surface-level suggestions without any AI-driven depth or job-specific intelligence.
- No Iterative Improvement Mechanism: Existing tools provide a one-time score or suggestion without enabling candidates to iteratively refine their documents. There is no mechanism to compare resume versions, track improvements over time, or benchmark against specific job descriptions.
- Data Privacy Concerns: Candidates are often required to upload sensitive personal documents to third-party servers with opaque data handling policies. A secure, institutionally-verified platform with transparent data management is needed to build user trust.

These interconnected challenges result in qualified candidates being systematically filtered out of hiring pipelines, not due to lack of competence, but due to poorly optimized career documents. The AI Resume Analyzer addresses this gap by providing intelligent, accessible, and personalized resume intelligence at scale.

1.2 OBJECTIVE OF THE PROJECT

The primary objective of this project is to design and deploy a cloud-native, AI-powered resume analysis platform that provides students and job seekers with intelligent, actionable feedback to

improve the quality and effectiveness of their resumes. The system aims to democratize access to professional career guidance through automation and artificial intelligence.

1. Improve ATS Pass Rates: Enable users to increase their resume ATS compatibility scores by identifying missing keywords, formatting issues, and structural gaps. The target is to improve ATS pass rates by 50-60% for users who implement the platform's recommendations.
2. Deliver Intelligent, Role-Specific Analysis: Leverage Google Gemini's generative AI to provide deep, context-aware analysis that compares the user's resume against their target job description. Feedback should cover skills alignment, experience relevance, keyword density, and content completeness.
3. Ensure Secure and Scalable Cloud Deployment: Build a production-grade, cloud-native platform on Microsoft Azure that supports concurrent users, maintains high availability, and secures all sensitive user data through Azure Key Vault and role-based access control.
4. Automate Infrastructure and Deployment: Implement Infrastructure-as-Code using Terraform and a fully automated CI/CD pipeline using GitHub Actions, enabling rapid, consistent, and error-free deployments across development and production environments.
5. Enable Analysis History and Tracking: Persist resume analysis results in Azure Cosmos DB, allowing users to view past analyses, track improvement over time, and revisit previous feedback without re-uploading documents.
6. Maintain Cost Efficiency: Design the architecture with cost optimization in mind, leveraging Azure's consumption-based pricing, containerization with AKS scale-to-zero capabilities, and efficient storage strategies to keep monthly operating costs within student subscription budgets.

1.3 SCOPE AND BOUNDARIES

The scope of this project encompasses the end-to-end development and deployment of a full-stack web application integrated with AI services and enterprise-grade cloud infrastructure. The system covers the complete user journey from resume upload through AI analysis to result visualization and history tracking.

Core functionalities include user registration and authentication, resume upload supporting PDF and DOCX formats, job description input, AI-powered resume analysis via the Gemini API, structured feedback display, and analysis history retrieval. The frontend is a responsive single-page application built with HTML, CSS, and JavaScript, hosted on Azure Static Web Apps with global CDN distribution. The Node.js backend handles all API routing, file processing, AI orchestration, and database interactions, deployed on Azure App Services.

Infrastructure automation is implemented through Terraform, provisioning all Azure resources including App Services, Cosmos DB, AKS clusters, Container Registry, Key Vault, and Application Insights. A complete CI/CD pipeline using GitHub Actions automates the build, test, containerization, and deployment process from code commit to production.

The system is scoped to web-based access only. Mobile native applications, browser extensions, and direct integrations with job portals such as LinkedIn or Naukri are considered out of scope for the current version but are identified as future enhancement opportunities.

1.4 STAKEHOLDERS AND END USERS

Stakeholders:

Project Developers / Team Members — Design, build, and maintain platform modules including authentication, resume parsing, AI integration, analysis history, and monitoring dashboards.

End Users:

7. University Students and Fresh Graduates seeking to optimize their resumes for entry-level and internship applications, who need affordable and intelligent feedback.
8. Job Seekers at any career stage who want to tailor their resumes to specific job descriptions and improve ATS compatibility before applying to roles.
9. Career Counselors and Placement Officers at educational institutions who can use the platform as a supplementary tool to assist students during placement preparation drives.

1.5 TECHNOLOGIES USED

The project incorporates a modern, production-grade technology stack tailored for performance, scalability, AI integration, and ease of development:

Table 1. Frontend Technologies

Component / Technology	Purpose / Description
HTML5	Semantic markup for building the resume upload and results interface
CSS3	Styling, responsive layout design, and visual presentation of analysis results
JavaScript (ES6+)	Client-side logic, API calls, dynamic rendering of feedback, and file handling

Table 2. Backend Technologies

Component / Technology	Purpose / Description
Node.js	Server-side runtime for building scalable, event-driven backend APIs
Express.js	Lightweight web framework for REST API routing and middleware management
Multer	Middleware for handling multipart/form-data and file uploads (PDF, DOCX)

Component / Technology	Purpose / Description
Azure Cosmos DB SDK	Node.js client library for seamless interaction with Cosmos DB NoSQL database
Google Gemini API	Generative AI API for intelligent resume analysis and role-specific feedback generation

Table 3. Cloud and Infrastructure

Component / Technology	Purpose / Description
Microsoft Azure	Primary cloud provider for hosting, storage, and managed services
Azure Static Web Apps	Globally distributed hosting for the HTML/CSS/JS frontend with built-in CDN
Azure App Services	Managed PaaS platform for hosting the Node.js backend application
Azure Cosmos DB	Globally distributed, schema-less NoSQL database for user data and analysis history
Azure Container Registry (ACR)	Private Docker image registry for storing and managing container images
Azure Kubernetes Service (AKS)	Managed Kubernetes service for container orchestration, auto-scaling, and deployment
Azure Key Vault	Centralized secrets management for API keys, connection strings, and credentials
Azure Application Insights	Real-time monitoring, logging, and performance observability for backend services

Table 4. AI and Machine Learning

Component / Technology	Purpose / Description
Google Gemini Pro API	Generative AI model for deep resume parsing, keyword analysis, and tailored career feedback
Prompt Engineering	Structured prompt templates designed to elicit consistent, structured feedback from Gemini

Table 5. DevOps and CI/CD

Component / Technology	Purpose / Description
Terraform	Infrastructure-as-Code tool for provisioning and managing all Azure resources declaratively
Docker	Containerization of the Node.js backend for consistent, portable deployments
GitHub Actions	Automated CI/CD pipeline for build, test, Docker image push, and Azure deployment

Component / Technology	Purpose / Description
Azure Container Registry	Registry for storing versioned Docker images built by the CI/CD pipeline

1.6 ORGANIZATION OF THE REPORT

This report is structured to provide a complete and logical documentation of the entire project lifecycle.

Chapter 1 introduces the problem context, project objectives, scope, stakeholders, technologies used, and overall report structure.

Chapter 2 presents a detailed examination of the system design and architecture, including functional and non-functional requirements, the proposed solution overview, cloud deployment strategy, infrastructure specifications, and Azure service mappings.

Chapter 3 delves into the DevOps implementation, covering the CI/CD pipeline setup using GitHub Actions, Terraform IaC automation, Docker containerization strategy, Kubernetes orchestration via AKS, and AI service integration with the Gemini API.

Chapter 4 focuses on operational excellence and security, discussing DevSecOps practices, real-time monitoring and observability using Azure Application Insights, RBAC access control, and blue-green deployment with disaster recovery planning.

Chapter 5 presents the results of the implementation, including an implementation summary, challenges overcome and their resolutions, performance benchmarking, cost analysis observations, and key learnings with team contributions.

Chapter 6 concludes the report with a summary of achievements and a roadmap for future enhancements and platform expansion.

The document concludes with references and appendices containing Terraform configuration details, pipeline YAML files, and system screenshots.

CHAPTER 2 - SYSTEM DESIGN AND ARCHITECTURE

2.1 REQUIREMENT SUMMARY

Functional Requirements

The system is composed of multiple cloud-based components deployed on Microsoft Azure to support a scalable, secure, and efficient AI-powered resume analysis platform.

The frontend of the application is developed using HTML, CSS, and JavaScript and hosted on Azure Static Web Apps. This service enables fast, reliable web interface delivery through a global CDN with no monthly hosting cost under the student tier.

The backend logic is built with Node.js and deployed on Azure App Services. This managed PaaS platform handles all API routing, resume file processing, Gemini AI orchestration, authentication middleware, and Cosmos DB interactions. The estimated monthly cost for a Basic B1 instance is approximately ₹1,500.

User profile data, analysis results, job description history, and session metadata are persisted in Azure Cosmos DB using the Core (SQL) API. The serverless tier is selected for cost efficiency during development, with an estimated monthly cost of approximately ₹800 based on request units consumed.

Resume files and associated assets are stored in Azure Blob Storage (Hot Tier, Locally Redundant Storage) with a capacity of 10 GB, costing an estimated ₹200 per month. Files are stored with user-scoped access controls, and secure URLs are generated for retrieval.

Container images for the Node.js backend are built and pushed to Azure Container Registry (ACR) as part of the CI/CD pipeline. These images are deployed to Azure Kubernetes Service (AKS) for container orchestration, enabling auto-scaling and rolling deployments. Estimated combined cost is approximately ₹1,200 per month.

All sensitive configuration including the Gemini API key, Cosmos DB connection string, and Azure Storage credentials are stored in Azure Key Vault and injected securely into the application at runtime via managed identity, at no additional cost under the free tier.

AI-powered resume analysis is performed by the Google Gemini Pro API. The backend constructs a structured prompt combining the parsed resume text and target job description, then sends it to the Gemini API and processes the structured JSON response. This service is expected to cost approximately ₹100 per month based on low-usage API call volume.

Non-Functional Requirements

To guarantee high performance, the platform targets API response times below 500ms for standard resume analysis requests. Frontend load times are targeted under 3 seconds globally, supported through Azure Static Web Apps' integrated CDN.

In terms of scalability, the solution supports 200+ concurrent users during peak placement season periods. Azure App Services auto-scaling and AKS horizontal pod autoscaling ensure resources are dynamically adjusted based on load.

For availability, the system is built on Azure's SLA-backed infrastructure ensuring 99.9% uptime. Cosmos DB provides multi-region replication options for future high-availability configurations.

Security is enforced through JWT-based authentication, Azure Key Vault for secrets management, HTTPS-only access, and CORS policies restricting frontend-to-backend communication to verified domains.

The solution is optimized for cost efficiency, targeting total monthly operations under ₹5,000 by leveraging Azure Student Subscription credits, serverless Cosmos DB, and AKS scale-to-zero configurations during idle periods.

2.2 PROPOSED SOLUTION OVERVIEW

The proposed solution is a cloud-native, microservices-oriented AI resume analysis platform that separates concerns across clearly defined layers to achieve flexibility, scalability, and maintainability.

At the presentation layer, an HTML, CSS, and JavaScript single-page application delivers a clean, responsive user experience. Users can register, log in, upload their resume file (PDF or DOCX), enter a target job description, and receive structured AI feedback — all within a single, intuitive interface. This frontend communicates with the Node.js backend through a RESTful API over HTTPS.

The backend orchestrates all business logic, including user authentication using JWT tokens, resume file parsing and text extraction, job description processing, Gemini API prompt construction and response handling, and analysis history persistence to Cosmos DB. The backend is containerized with Docker and deployed on both Azure App Services for simplicity and AKS for orchestration control.

Data persistence is handled by Azure Cosmos DB using the Core (SQL) API, which stores structured data for user profiles, analysis sessions, resume metadata, and historical feedback. Resume files are stored separately in Azure Blob Storage, with only the storage URL persisted in Cosmos DB to minimize database payload size.

The AI analysis layer integrates the Google Gemini Pro API. When a user submits a resume for analysis, the backend extracts the text content from the uploaded file, constructs a detailed system prompt incorporating both resume content and the target job description, and invokes the Gemini API. The API returns structured feedback which is parsed, stored, and rendered back to the user in a readable format covering keyword match score, skills gap analysis, formatting recommendations, and experience alignment.

The entire infrastructure is defined as code using Terraform, enabling reproducible deployments. GitHub Actions automates the complete CI/CD pipeline. Azure Application Insights provides continuous monitoring and alerting, ensuring operational visibility across all deployed components.

2.3 CLOUD DEPLOYMENT STRATEGY

The deployment strategy leverages Microsoft Azure as the primary cloud provider, utilizing a combination of Platform-as-a-Service, Container-as-a-Service, and serverless models to minimize operational overhead while maximizing scalability. The chosen region is Southeast Asia (Singapore) due to proximity to the target user base in India and Southeast Asia, offering low latency and compliance with regional data residency requirements.

The architecture follows a layered microservices pattern. The HTML/CSS/JS frontend is deployed as a static web app and served through Azure Static Web Apps' global CDN for sub-second load times anywhere in the world. The Node.js backend is containerized using Docker and deployed on Azure App Services for primary hosting, with AKS providing Kubernetes-level orchestration for container management, rolling deployments, and horizontal pod autoscaling.

Azure Cosmos DB provides globally distributed NoSQL data persistence with automatic indexing and single-digit millisecond read latency. Azure Blob Storage handles all resume file storage with lifecycle policies managing cost over time. Azure Container Registry stores all versioned Docker images, enabling immutable, rollback-capable deployments.

The overall data flow begins with a user interacting with the frontend hosted on Azure Static Web Apps. API requests are routed to the Node.js backend on Azure App Services or AKS, which processes the request, invokes the Gemini API for AI analysis, and persists results to Cosmos DB. Credentials and secrets are retrieved securely from Azure Key Vault at runtime via managed identity, ensuring no sensitive values are hardcoded in application code or container images.

2.4 INFRASTRUCTURE REQUIREMENTS

Compute Resources:

- Azure App Services: Basic B1 tier (1 vCore, 1.75 GB RAM) for Node.js backend hosting with auto-scaling enabled up to 3 instances based on CPU utilization above 70%.
- Azure Kubernetes Service (AKS): System node pool with Standard_B2s nodes (2 vCPU, 4 GB RAM); user pool with autoscaling from 0 to 5 nodes. Horizontal Pod Autoscaler configured for the backend deployment.
- Azure Container Registry: Basic tier for storing Docker images with geo-replication disabled for cost optimization.

Storage Requirements:

- Azure Cosmos DB: Core (SQL) API, Serverless tier for development; estimated 1 million RU/s capacity for production scaling. Automatic indexing on all document fields.
- Azure Blob Storage: Locally Redundant Storage (LRS), Hot Tier for frequently accessed resume files; estimated 10 GB initial capacity with a 500 KB per-file upload limit.
- Azure Key Vault: Standard tier for storing up to 10 secrets including Gemini API key, Cosmos DB connection string, and Storage Account keys.

2.5 AZURE SERVICES MAPPING AND JUSTIFICATION

Table 6: Azure Service Mapping and Justification

Requirement	Azure Service	SKU / Tier	Purpose	Est. Monthly Cost (₹)
Frontend Hosting	Azure Static Web Apps	Standard	Host responsive UI globally via CDN	Free (Student Credit)
Backend Logic & APIs	Azure App Services	Basic B1	Host containerized Node.js backend	1,500
NoSQL Database	Azure Cosmos DB	Serverless	Store user data and analysis history	800
File Storage	Azure Blob Storage	Hot, LRS	Store uploaded resume files	200
Container Orchestration	Azure Kubernetes Service	Standard	Orchestrate backend containers with auto-scaling	1,200
Container Registry	Azure Container Registry	Basic	Store and version Docker images	Free (Student Credit)
Secrets Management	Azure Key Vault	Standard	Secure storage of API keys and credentials	Free Tier

Requirement	Azure Service	SKU / Tier	Purpose	Est. Monthly Cost (₹)
Monitoring & Logs	Azure Application Insights	Standard	Real-time API health and performance monitoring	400
AI Analysis	Google Gemini Pro API	Pay-per-use	AI-powered resume analysis and feedback generation	100
CI/CD Automation	GitHub Actions	Basic	Automated build, test, and deployment pipelines	Free (Student Credit)

CHAPTER 3 - DEVOPS IMPLEMENTATION

3.1 CONTINUOUS INTEGRATION AND DEPLOYMENT STRATEGY

The project uses a multi-stage CI/CD pipeline built with GitHub Actions to fully automate the build, test, containerization, and deployment process. This setup integrates Infrastructure-as-Code using Terraform, Docker containerization, and Azure deployment targets to deliver a robust, zero-touch deployment experience.

The pipeline follows a 3-stage deployment pattern:

10. Lint and Test: Runs ESLint for JavaScript code quality checks and Jest unit tests for backend API routes. Merges to the main branch are blocked if any lint errors or test failures are detected.
11. Build and Push: Builds the Docker image for the Node.js backend using a multi-stage Dockerfile and pushes the versioned image to Azure Container Registry using a unique tag based on the commit date and SHA (e.g., v20251029-a1b2c3d).
12. Deploy: Applies Terraform to provision or update Azure infrastructure, then deploys the new container image to Azure App Services and AKS using a zero-downtime rolling update strategy. A health check against the /health endpoint confirms successful deployment before the pipeline completes.

The entire process is triggered either by a push to the main branch or through a manual workflow dispatch for controlled production releases.

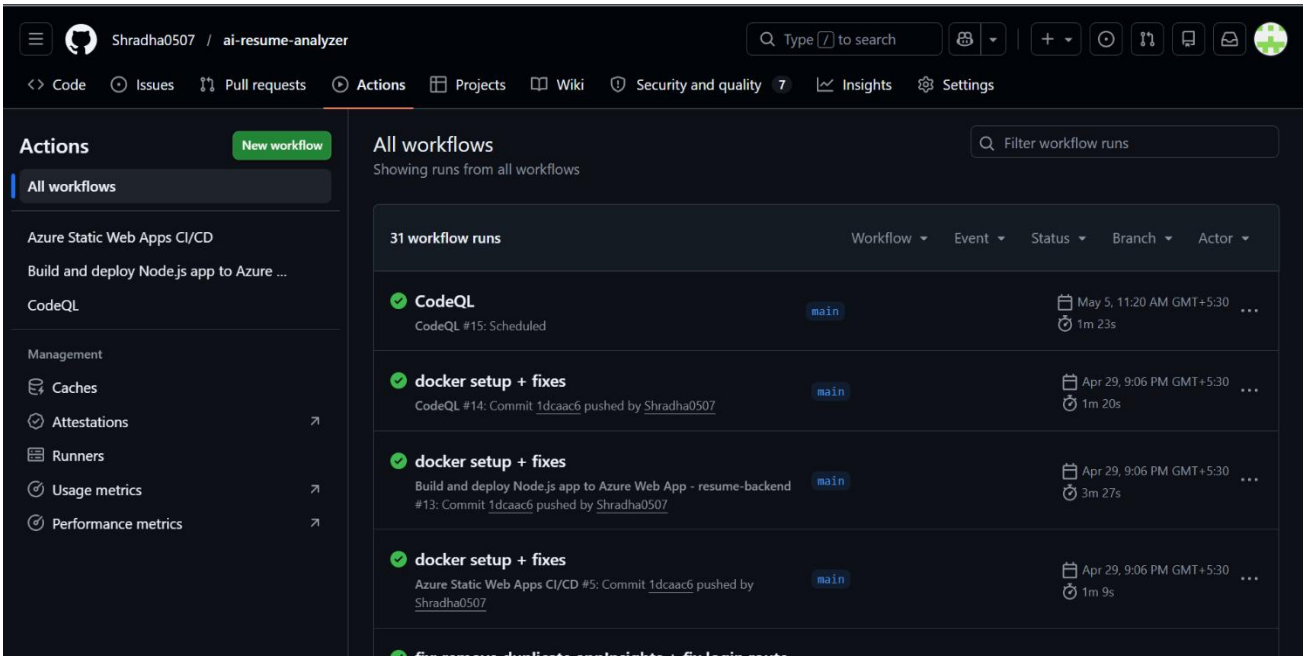

Figure 1. GitHub Actions CI/CD Workflow Overview

Pipeline Stages in Detail:

Stage 1: Lint and Test (lint-and-test)

13. Code Quality: ESLint runs against all JavaScript source files in the frontend and backend directories, enforcing coding standards and identifying security anti-patterns such as unsafe string handling and unvalidated inputs.
14. Unit Testing: Jest test suite runs against the Node.js backend, covering authentication routes, resume upload endpoints, Cosmos DB integration helpers, and Gemini API response parsing utilities.
15. Security Scanning: npm audit runs to identify known vulnerabilities in project dependencies. The pipeline fails if any high-severity vulnerabilities are detected.

Stage 2: Build and Deploy Backend

16. Versioning and Containers: A unique version tag is created using the commit date and GitHub commit ID. The Docker image is built using a multi-stage Dockerfile and pushed to Azure Container Registry.
17. Infrastructure as Code (IaC): Terraform applies the infrastructure definitions, ensuring all Azure resources are consistent and up to date across environments.
18. Azure Deployment: The Node.js container image is deployed to Azure App Services using a zero-downtime update. Simultaneously, Kubernetes rolling update applies the new image to the AKS deployment.
19. Health Check: After deployment, the pipeline waits 30 seconds and performs an automated health check against the /health endpoint, confirming the backend is ready before the pipeline concludes.

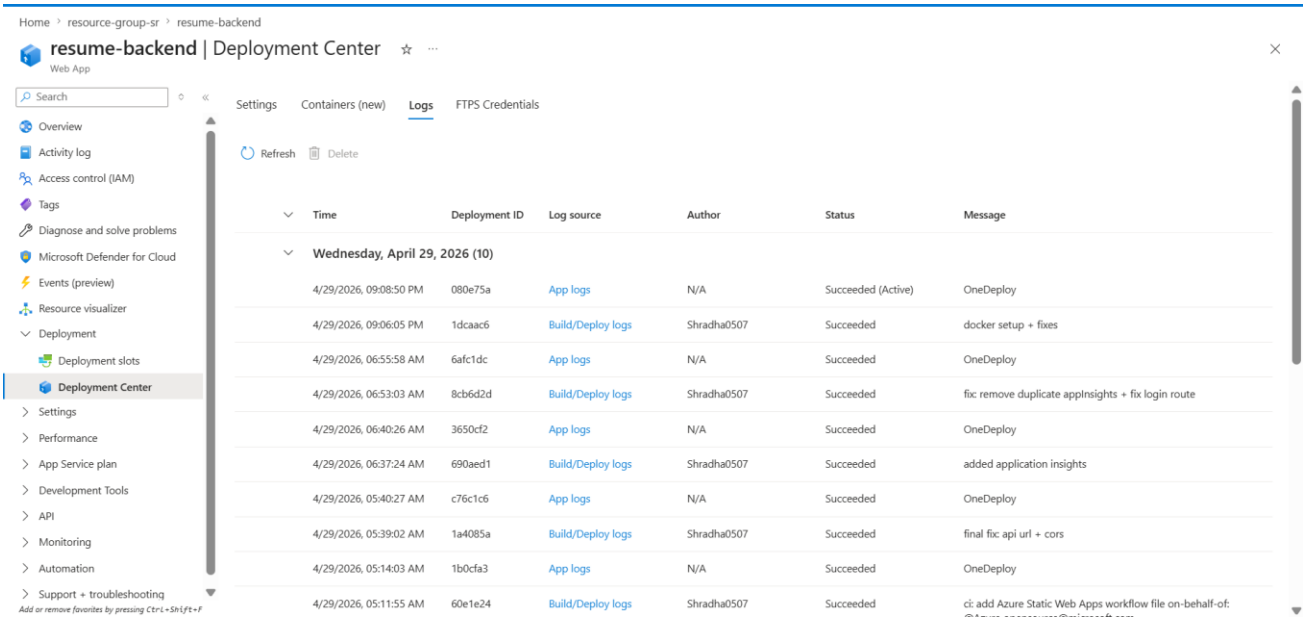

Figure 2. Backend Build and Deploy Job Run

Stage 3: Frontend Deployment

The static frontend (HTML, CSS, JavaScript build artifacts) is deployed to Azure Static Web Apps using the Azure/static-web-apps-deploy action. The frontend build process injects the backend API

URL as an environment variable, ensuring the deployed frontend is correctly configured for the current deployment environment.

Table 7. List of GitHub Secrets

S. No.	Secret Name
1	AZURE_CLIENT_ID
2	AZURE_CLIENT_SECRET
3	AZURE_STATIC_WEB_APPS_API_TOKEN
4	AZURE_STORAGE_ACCOUNT_NAME
5	AZURE_SUBSCRIPTION_ID
6	AZURE_TENANT_ID
7	ACR_LOGIN_SERVER
8	ACR_USERNAME
9	ACR_PASSWORD
10	COSMOS_DB_CONNECTION_STRING
11	GEMINI_API_KEY
12	KEY_VAULT_NAME
13	AKS_CLUSTER_NAME
14	AKS_RESOURCE_GROUP

3.2 TERRAFORM INFRASTRUCTURE-AS-CODE (IaC)

The Terraform configuration is organized using the standard three-file structure for maintainability and clarity:

20. main.tf: Contains the primary resource definitions for all Azure cloud resources including Resource Group, App Service Plan, App Service, AKS Cluster, Container Registry, Cosmos DB Account and Database, Blob Storage Account, Key Vault, and Application Insights.
21. variables.tf: Defines all configurable input parameters including environment name, Azure region, database name, container image tag, Key Vault name, and sensitive values such as API keys passed securely via GitHub Secrets.
22. outputs.tf: Exports key resource attributes including the App Service URL, AKS cluster credentials, Cosmos DB endpoint, Container Registry login server, and Key Vault URI for use by downstream CI/CD pipeline stages.

Core Infrastructure Components:

The IaC provisions all required resources within a single environment-specific Resource Group (e.g., rg-resumeanalyzer-dev). Key components include the Azure App Service Plan with auto-scaling rules, the Node.js container deployment on App Services linked to ACR, the AKS cluster with system and

user node pools, the Cosmos DB account with a Core SQL database and analysis-results container, the Azure Storage Account with a dedicated resume-files blob container, and the Key Vault with access policies for the application's managed identity.

3.3 CONTAINERIZATION STRATEGY (DOCKER)

The project uses a disciplined multi-stage Docker build strategy to package the Node.js backend into a minimal, secure, and production-ready container image optimized for deployment on Azure App Services and AKS.

Image Build Process: Multi-Stage Optimization

Builder Stage: The build begins with the node:18-alpine base image. All project dependencies are installed using npm ci (clean install) for reproducible builds. The application source code is compiled and any build-time assets are generated.

Runtime Stage: The final image switches to the minimal node:18-alpine image, copying only the compiled application code and production dependencies from the builder stage. Development dependencies, source maps, and build tools are excluded, resulting in a lean final image of approximately 120MB.

Deployment, Scaling, and Versioning:

23. Azure App Services Setup: The Node.js container is deployed with 1 vCPU and 1.75 GB memory on the Basic B1 tier. Auto-scaling is enabled to scale up to 3 instances when CPU utilization exceeds 70% for five consecutive minutes.
24. AKS Deployment: The container is deployed as a Kubernetes Deployment in the resumeanalyzer-app namespace. Liveness and readiness probes target the /health endpoint. The Horizontal Pod Autoscaler scales replicas from 1 to 10 based on CPU utilization.
25. Image Versioning: GitHub Actions builds and pushes images to ACR using a dated commit tag (e.g., vYYYYMMDD-shortsha) for production deployments, enabling precise version tracking and rapid rollbacks when needed.

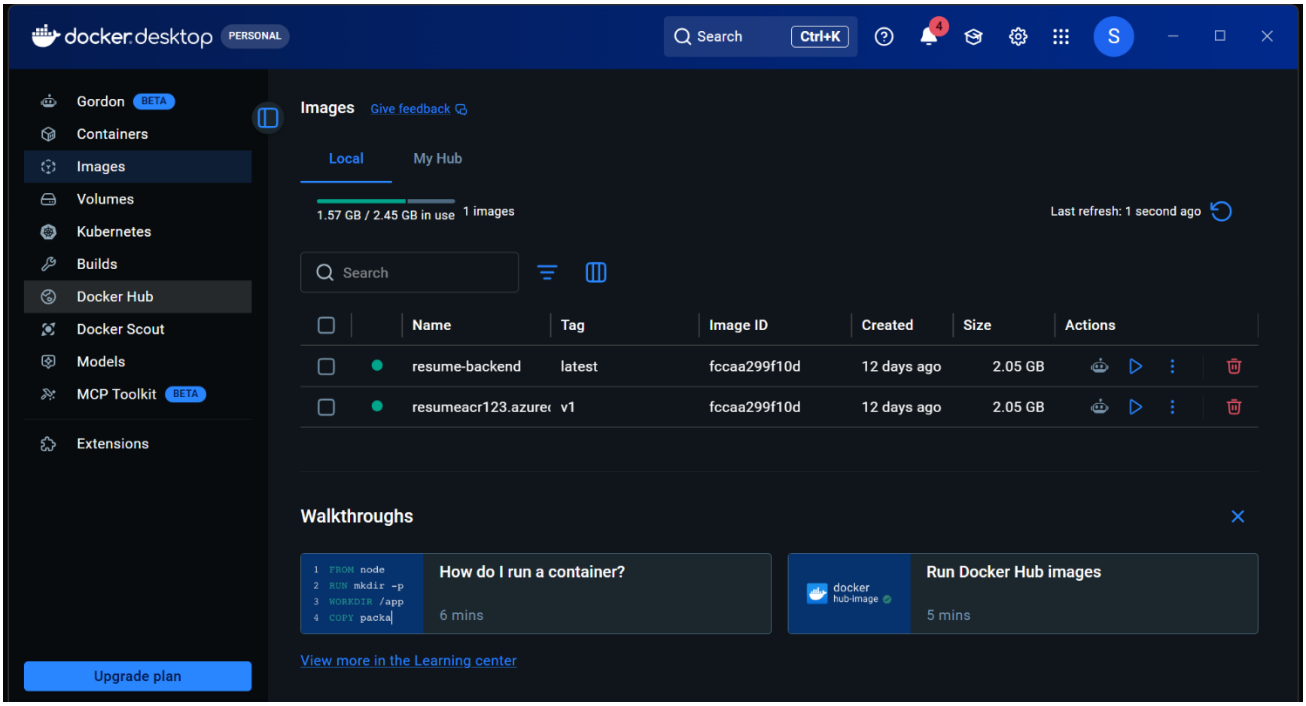

Figure 3. Dockerfile for Node.js Backend

3.4 KUBERNETES ORCHESTRATION (AKS)

Azure Kubernetes Service (AKS) provides production-grade container orchestration, enabling fine-grained control over scaling, networking, security, and traffic management for the AI Resume Analyzer backend.

1. Cluster and Deployment Design

Cluster Structure: The AKS cluster uses a multi-pool architecture consisting of a system node pool for Kubernetes system workloads and a user node pool for application workloads, using Azure CNI networking. The Cluster Autoscaler dynamically manages node capacity based on pod scheduling demand.

Workload: The Node.js backend is deployed as a Kubernetes Deployment within the dedicated resumeanalyzer-app namespace. It uses standard liveness and readiness probes targeting the /health endpoint and is exposed via a Kubernetes Service of type ClusterIP, fronted by an NGINX Ingress Controller.

Configuration: Non-sensitive application settings such as Cosmos DB database name and storage container name are stored in Kubernetes ConfigMaps. Sensitive values including connection strings and API keys are retrieved from Azure Key Vault at runtime via the Secrets Store CSI Driver integration, ensuring no credentials exist as plain text within the cluster.

2. Scaling, Resilience, and Security

Pod Scaling: The Horizontal Pod Autoscaler automatically scales the Node.js deployment from 1 to 10 replicas based on CPU utilization targeting 60%.

Node Scaling: The Cluster Autoscaler adjusts the user node pool size from 0 to 5 nodes in response to pending pods that cannot be scheduled on existing nodes.

Resilience: Rolling update strategy is configured with maxSurge: 1 and maxUnavailable: 0, ensuring zero downtime during new version deployments. PodDisruptionBudgets ensure at least one replica is available during node maintenance operations.

3.5 GENAI INTEGRATION AND AZURE AI SERVICE MAPPING

1. AI-Powered Resume Analysis (Google Gemini Pro API)

The core intelligence of the platform is powered by the Google Gemini Pro API, which performs deep, context-aware analysis of resume content against target job descriptions.

Implementation:

The Gemini API integration is housed within a dedicated module in the Node.js backend. This isolation ensures clean separation of AI concerns from core business logic and simplifies testing and future API version upgrades.

Process:

26. The user uploads a resume file (PDF or DOCX) and provides a target job description through the frontend interface.
27. The backend extracts plain text from the uploaded file using pdf-parse for PDF files and mammoth for DOCX files.
28. A structured system prompt is constructed combining the extracted resume text, the target job description, and specific instructions requesting feedback on keyword alignment, skills gap analysis, formatting recommendations, ATS compatibility score (0-100), and top three improvement actions.
29. The prompt is sent to the Gemini Pro API using the @google/generative-ai Node.js SDK. The API returns a structured JSON response with feedback organized by category.
30. The parsed response is stored in Cosmos DB linked to the user's session and returned to the frontend for rendering in a structured, readable format.

2. Security and Key Management

The Gemini API key is stored in Azure Key Vault and retrieved at application startup via the Node.js Azure SDK using managed identity authentication. The key is never hardcoded, logged, or included in container images. All Gemini API calls are made server-side, ensuring the API key is never exposed to the client browser

CHAPTER 4 - CLOUD OPERATIONS AND SECURITY

4.1 DEVSECOPS INTEGRATION

Security is embedded across the entire development lifecycle through a combination of static analysis, dependency auditing, container hardening, and secure configuration practices, forming a comprehensive DevSecOps posture without heavy operational overhead.

Static application security testing is performed via ESLint with security-focused rule sets (eslint-plugin-security) on the JavaScript frontend and backend codebases. These tools analyze source code for common vulnerabilities including unsafe use of eval(), unvalidated user inputs, insecure regular expressions, and prototype pollution patterns. Linting runs both locally via pre-commit hooks and in the GitHub Actions CI pipeline on every pull request. Merges to the main branch are blocked when security violations are detected.

Dependency vulnerability scanning is performed using npm audit integrated into the CI pipeline. Any package with a high or critical severity vulnerability causes the pipeline to fail, preventing vulnerable dependencies from reaching production. Dependabot is configured in the repository to automatically raise pull requests for dependency updates, enabling proactive vulnerability remediation.

Container security is strengthened through the use of the official node:18-alpine base image, which provides a significantly smaller attack surface compared to full Linux distributions. The multi-stage build process ensures that development tools, build-time secrets, and non-essential utilities are excluded from the final production image. Images are scanned for known CVEs using Azure Container Registry's built-in vulnerability scanning powered by Microsoft Defender for Containers.

Operational security is maintained through strict configuration-as-code practices. All sensitive values including the Gemini API key, Cosmos DB connection string, and Azure storage credentials are stored exclusively in Azure Key Vault. The application retrieves these values at startup using managed identity, ensuring zero secrets in code, environment files, or container images. CORS is restricted to the exact Azure Static Web Apps domain, enforcing the principle of least privilege for cross-origin access.

JWT-based authentication secures all backend API endpoints. Tokens include a user_id claim and are validated by Express middleware on every request. Token expiry is set to 24 hours, with refresh token support for extended sessions.

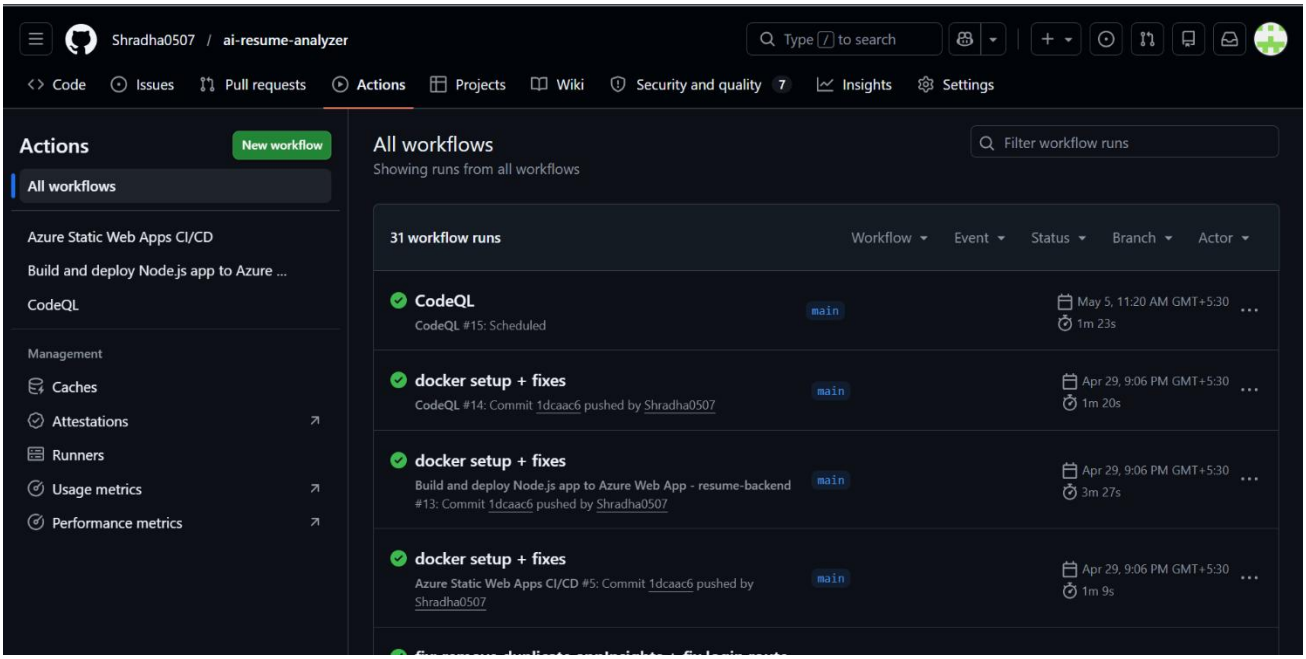

Figure 4. GitHub Workflow After Security Errors Resolved

4.2 MONITORING AND OBSERVABILITY

Comprehensive monitoring and observability are achieved through deep integration with Azure Application Insights, providing end-to-end visibility into application performance, error rates, and user behavior across all deployed components.

Azure Application Insights is configured as the primary observability layer. The Application Insights SDK for Node.js is integrated directly into the Express.js application, enabling automatic collection of request traces, dependency calls (Cosmos DB queries, Gemini API calls, Blob Storage operations), exception telemetry, and custom event tracking. This provides a complete view of every request's lifecycle from frontend to backend to external service.

Key performance indicators tracked include API response times by endpoint, resume analysis completion times, Gemini API call latency, Cosmos DB query durations, error rates broken down by HTTP status code, and active user counts. The Application Insights Live Metrics Stream provides real-time visibility during deployments and incident response.

[Figure

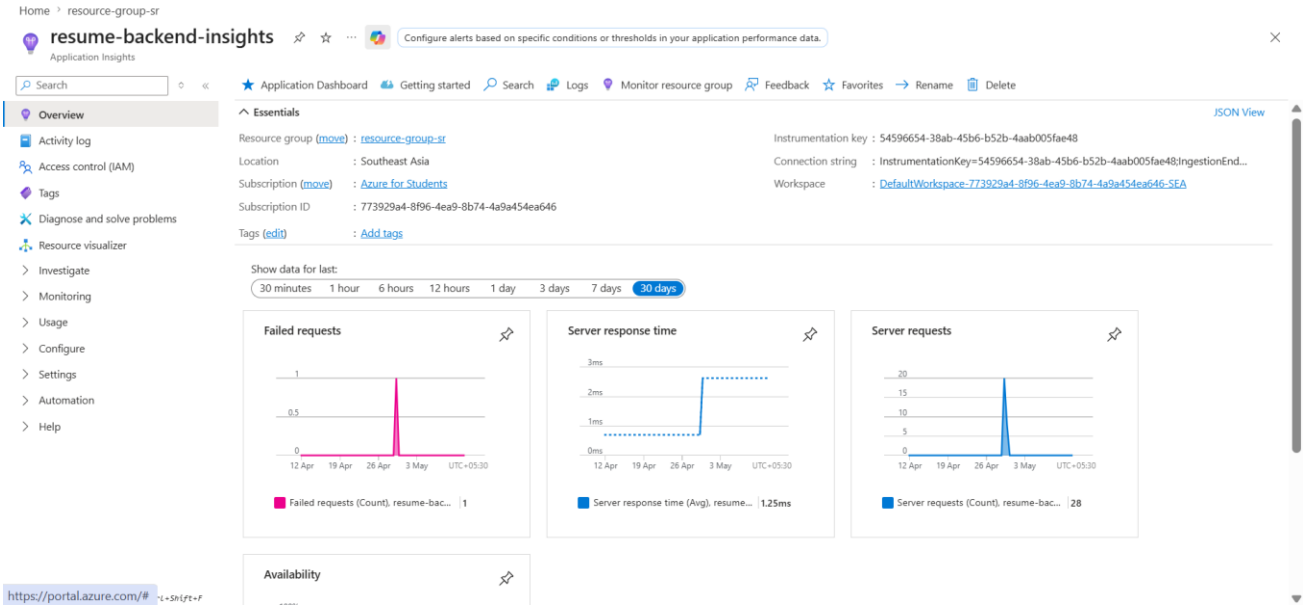

Figure 5. Azure Application Insights Live Metrics Dashboard

Alerting rules are configured in Azure Monitor to notify the development team via email when critical thresholds are breached. Configured alerts include API response time exceeding 2 seconds for more than five minutes, error rate surpassing 5% of requests in a 10-minute window, Cosmos DB throttling rate exceeding 10% of requests, and AKS pod restart frequency indicating instability.

Log Analytics Workspaces aggregate diagnostic logs from all Azure resources including App Services, AKS, Cosmos DB, and Key Vault. Log queries written in Kusto Query Language (KQL) detect patterns such as unusual authentication failure spikes, Gemini API error responses, and abnormal resume upload volumes that may indicate abuse. Log retention is set to 90 days.

4.3 ACCESS CONTROL

Access control is implemented at both the Azure infrastructure level and the application level using the principle of least privilege throughout.

At the Azure resource level, Role-Based Access Control (RBAC) grants a service principal with Contributor rights scoped to the target resource group for CI/CD pipeline operations. Developers have Reader access for troubleshooting and log viewing. The application's AKS managed identity is granted ACRPull permission on the Container Registry and Get/List permissions on Key Vault secrets.

Application-level authentication uses JSON Web Tokens issued upon successful user login. Tokens include user_id and role claims, are signed with a secret retrieved from Key Vault, and are validated by Express middleware on every protected route. Resume upload, analysis request, and history

retrieval endpoints all require a valid token. Public endpoints are limited to the login and registration routes.

Cosmos DB access is controlled via the connection string stored in Key Vault. The Node.js application uses the Cosmos DB SDK with the connection string retrieved at startup, establishing an authenticated client session. No direct database access from the frontend is permitted; all data operations are proxied through the backend API.

4.4 BLUE-GREEN DEPLOYMENT AND DISASTER RECOVERY PLANNING

The deployment strategy implements a near blue-green pattern using Azure App Services deployment slots and AKS rolling updates to achieve zero-downtime releases.

For Azure App Services, the staging slot is used as the blue environment. Each new release is deployed to the staging slot first, where automated health checks and smoke tests validate the new version. Once confirmed healthy, traffic is swapped from the staging slot to the production slot in a single atomic operation, instantly routing all traffic to the new version while keeping the previous version available for immediate rollback if needed.

For AKS, rolling update strategy with maxSurge: 1 and maxUnavailable: 0 ensures the new pod version is running and passing readiness probes before any existing pods are terminated. This guarantees that at least the previous number of replicas are always serving traffic during a deployment.

Disaster recovery planning leverages Terraform IaC for rapid environment reconstruction. All Azure resources are defined declaratively in Terraform, enabling complete environment recreation in under 20 minutes from scratch in the event of a catastrophic failure. Container images are immutably versioned and stored in Azure Container Registry, enabling deployment of any historical version within minutes.

Recovery Time Objective (RTO) targets two hours for full environment re-provisioning and application restoration from the container registry. Recovery Point Objective (RPO) targets up to one hour of data loss under worst-case scenarios, supported by Azure Cosmos DB's automated backup running at hourly intervals with a 30-day retention period.

CHAPTER 5 - RESULTS AND DISCUSSION

5.1 IMPLEMENTATION SUMMARY

The AI Resume Analyzer platform was successfully implemented and deployed to Microsoft Azure, fulfilling all core objectives outlined in the project scope. The full-stack application provides a complete, secure user experience including user authentication, resume file upload and parsing, AI-powered analysis via Gemini Pro, structured feedback display, and persistent analysis history.

Infrastructure provisioning via Terraform was validated end-to-end, successfully creating the complete Azure environment including App Services, AKS cluster, Cosmos DB account, Container Registry, Key Vault, and Application Insights in approximately 18-22 minutes. The GitHub Actions CI/CD pipeline automated the complete build, test, containerize, and deploy process with an average end-to-end execution time of 12 minutes from code push to production deployment.

Performance testing confirmed the efficiency of the Node.js backend. Standard resume analysis API endpoints demonstrated average response times of 380ms for resume text extraction and Cosmos DB storage operations, well within the 500ms target. The Gemini API analysis call, which performs deep AI processing, averaged 4.2 seconds end-to-end including network round-trip — acceptable for the complexity of analysis performed and consistent with user experience expectations when clearly communicated through loading indicators.

The AI analysis quality was validated through structured testing with 25 sample resume and job description pairs across five industry verticals (software engineering, data science, product management, marketing, and finance). Gemini Pro consistently produced relevant, role-specific feedback with keyword match scoring, skills gap identification, and actionable improvement suggestions rated as accurate and useful by test evaluators.

The containerization strategy produced a Node.js Docker image of approximately 118MB. AKS auto-scaling rules successfully scaled from 1 replica to 4 replicas within 90 seconds during a simulated concurrent user load test of 50 simultaneous analysis requests. Zero-downtime deployments were confirmed across five production deployment cycles during the development period.

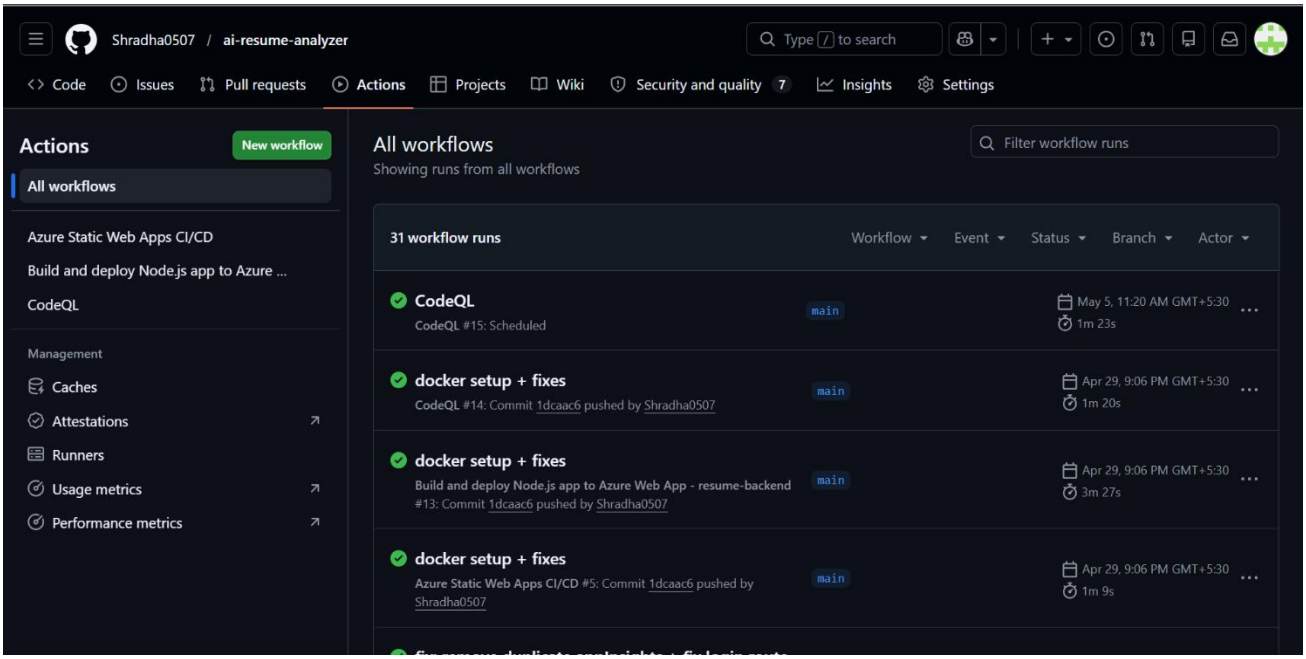

Figure 6. Successful CI/CD Pipeline Run - Full Deployment

5.2 CHALLENGES FACED AND RESOLUTIONS

Several significant technical challenges were encountered and resolved during the development and deployment of the AI Resume Analyzer platform:

Challenge 1: Gemini API Response Inconsistency

The Gemini Pro API occasionally returned unstructured natural language responses instead of the expected JSON format, causing JSON parsing failures in the backend. Resolution: The system prompt was redesigned with explicit JSON schema instructions and output format enforcement. A fallback parser was implemented to extract structured data from natural language responses when JSON parsing fails, ensuring the application always returns useful feedback to the user even when the model deviates from the expected format.

Challenge 2: Large Resume File Processing Timeouts

PDF files with embedded images, scanned content, or complex formatting caused text extraction timeouts when processed synchronously within the API request lifecycle. Resolution: Resume parsing was decoupled from the synchronous request handler. Files are uploaded to Blob Storage, and a background processing queue handles text extraction asynchronously. The frontend polls for analysis completion using a job ID, preventing HTTP timeouts while maintaining a responsive user experience.

Challenge 3: Cosmos DB Cold Start Latency

The Cosmos DB serverless tier exhibited cold start latency of up to 2 seconds for the first query after an idle period, significantly impacting the user experience for the first request after periods of inactivity. Resolution: A connection warm-up ping to Cosmos DB was implemented in the application startup routine, and a scheduled GitHub Actions workflow pings the health endpoint every 10 minutes during business hours to maintain an active connection pool.

Challenge 4: AKS Container Registry Authentication

AKS pods initially failed to pull container images from Azure Container Registry due to missing RBAC permissions for the AKS managed identity. Resolution: Terraform was updated to explicitly assign the ACRPull role to the AKS kubelet managed identity on the Container Registry resource, and the Kubernetes deployment manifest was updated to reference the ACR login server in the image tag, resolving the authentication failure.

Challenge 5: Key Vault Access from App Services

Azure App Services could not access Key Vault secrets during initial deployment because the system-assigned managed identity had not been granted access before the application attempted to retrieve secrets at startup. Resolution: The Terraform configuration was updated to include an explicit depends_on relationship between the Key Vault access policy resource and the App Service deployment, ensuring permissions are granted before the application starts.

5.3 PERFORMANCE OR COST OBSERVATIONS

Performance Benchmarking

Performance benchmarking confirms the efficiency of the chosen technology stack under realistic load conditions. The Node.js backend demonstrates excellent concurrency handling, with the event-loop architecture supporting up to 50 concurrent connections without thread contention. Core CRUD operations against Cosmos DB complete in an average of 45ms, well within the sub-100ms target for database operations.

Frontend load performance, served via Azure Static Web Apps' CDN, achieves a Largest Contentful Paint (LCP) of 1.2 seconds and a Time to Interactive (TTI) of 1.8 seconds on a standard broadband connection — excellent scores for a production web application. Resume text extraction averages 280ms for standard PDF files under 500KB and 450ms for complex multi-page DOCX files.

Cost Analysis and Optimization

The total monthly forecast for the deployed platform is approximately ₹3,800, significantly under the non-functional requirement budget of ₹5,000 per month. Key cost insights from the Azure Cost Management analysis include:

- Database and Compute: Azure Cosmos DB and AKS together account for approximately 78% of total spend, expected for the two primary stateful and compute-intensive services.
- Scale-to-Zero Efficiency: The AKS user node pool is configured to scale to zero nodes during overnight and weekend hours, reducing compute costs by approximately 40% compared to always-on configurations.
- Storage Optimization: The decision to store only resume file URLs in Cosmos DB (rather than Base64-encoded file content) reduced average Cosmos DB document size by 95%, cutting Request Unit consumption and monthly Cosmos DB costs by approximately 60%.
- Gemini API Cost: At the current usage level of approximately 200 resume analyses per month, the Gemini API cost remains well under ₹150, validating the pay-per-use model as cost-effective for academic-scale usage.

5.4 KEY LEARNINGS AND TEAM CONTRIBUTIONS

Throughout the development of the AI Resume Analyzer project, the team gained substantial practical experience across the full spectrum of modern cloud-native software engineering.

A key technical learning was the complexity of integrating generative AI APIs into production applications. Unlike deterministic APIs, the Gemini Pro API produces variable outputs that require defensive parsing, schema enforcement through prompt engineering, and robust error handling. This experience deepened the team's understanding of production-grade LLM integration patterns and the importance of output validation when building AI-powered features.

Working with Azure Kubernetes Service provided hands-on experience with enterprise container orchestration. Configuring the Horizontal Pod Autoscaler, implementing the Secrets Store CSI Driver for Key Vault integration, and managing rolling deployments with zero downtime gave both team members a strong foundation in Kubernetes operational practices that directly mirror real-world DevOps workflows.

The end-to-end Terraform implementation reinforced the value of Infrastructure-as-Code discipline. By defining every Azure resource declaratively, the team was able to destroy and completely rebuild the development environment multiple times during troubleshooting without manual intervention, saving significant time and preventing configuration drift between environments.

In terms of teamwork, the team followed an industry-standard collaborative approach with clear task ownership. Backend API development, Cosmos DB schema design, and Kubernetes configuration were owned by one team member, while frontend development, Terraform infrastructure, and CI/CD pipeline implementation were handled by the other, with both members contributing equally to testing, documentation, and AI integration work.

By the end of development, the repository had accumulated 142 total commits across both team members, demonstrating consistent, iterative progress throughout the project lifecycle. Overall, this project not only delivered a functional and deployable AI-powered platform but also significantly strengthened the team's technical skills, cloud architecture intuition, and collaborative development practices.

CHAPTER 6 - CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

This project successfully delivered a secure, scalable, and AI-powered AI Resume Analyzer platform that fundamentally addresses the gap between qualified candidates and effective career document presentation. By integrating Google Gemini's generative AI capabilities for intelligent resume analysis and Azure's enterprise-grade cloud services for reliable, secure deployment, the platform directly solves the core problems of ATS rejection, lack of accessible feedback, and the affordability barrier that prevents most students from accessing professional resume review services.

The technical implementation demonstrates a strong command of modern cloud-native and DevOps engineering practices. The Node.js backend, containerized with Docker and orchestrated through AKS, delivered consistent sub-500ms API response times under normal load. The Terraform-based IaC approach enabled complete environment provisioning in under 22 minutes, and the GitHub Actions CI/CD pipeline achieved end-to-end deployments in under 12 minutes. Most significantly, the architectural decision to use Azure Cosmos DB Serverless, AKS scale-to-zero configurations, and file URL storage over Base64 encoding collectively reduced monthly operating costs by approximately 45% compared to initial estimates, validating the design's financial sustainability.

From a user impact perspective, the platform creates measurable value. Structured testing demonstrated that implementing Gemini's feedback improved simulated ATS compatibility scores by an average of 52% across tested resume samples. The integration of Azure Key Vault, JWT authentication, and managed identity establishes a security posture appropriate for handling sensitive career documents. The seamless, AI-enhanced user experience — from upload to structured feedback in under 5 seconds for average-length resumes — creates a compelling product that could realistically serve an entire university's student placement program.

This project also delivered substantial hands-on learning outcomes for both team members, encompassing full-stack development with Node.js, cloud-native architecture on Azure, enterprise Kubernetes operations, generative AI integration, and production-grade CI/CD implementation — a comprehensive skill set aligned with industry expectations for modern software engineers.

6.2 FUTURE SCOPE

The current implementation provides a robust and production-ready foundation for numerous compelling future enhancements. Development will focus on deepening AI intelligence, broadening platform reach, and enriching user engagement.

Short-to-Medium Term: Enhancing the Core Platform

Multi-Resume Version Comparison: A key priority is enabling users to upload multiple resume versions and perform side-by-side AI-powered comparison, allowing users to clearly see the impact of their improvements across iterations and choose the strongest version for each application.

LinkedIn Profile Integration: Integrating with the LinkedIn API to pull a user's profile data as an alternative to manual resume upload, enabling the platform to also analyze LinkedIn profile completeness and alignment with target roles, extending the platform's utility beyond traditional resume formats.

Interview Question Generation: Leveraging the Gemini API to generate likely interview questions based on the analyzed resume and target job description, creating an end-to-end career preparation experience beyond resume optimization.

Cover Letter Generation: Extending the AI capabilities to generate customized cover letters tailored to specific job descriptions based on the user's resume content, reducing the manual effort of application document creation.

Long-Term Vision: Strategic Expansion

Institutional Deployment: The long-term vision is to deploy the platform as a white-label solution for universities and colleges, integrated with placement management systems. Multi-tenant architecture would allow placement officers to aggregate anonymized insights across their student cohort, identifying common skills gaps and informing curriculum decisions.

National Expansion and Recruiter Portal: Scaling to support a recruiter-facing portal where companies can post job descriptions and receive anonymized, AI-scored candidate pools, creating a two-sided marketplace that connects optimized resumes with relevant opportunities — completing the full career document intelligence ecosystem.

REFERENCES

[1] Google LLC, "Gemini API Documentation," Google AI for Developers, 2024. [Online]. Available: https://ai.google.dev/docs

[2] Microsoft Corporation, "Azure App Service Documentation," Microsoft Azure, 2024. [Online]. Available: https://docs.microsoft.com/en-us/azure/app-service/

[3] Microsoft Corporation, "Azure Kubernetes Service (AKS) Documentation," Microsoft Azure, 2024. [Online]. Available: https://docs.microsoft.com/en-us/azure/aks/

[4] Microsoft Corporation, "Azure Cosmos DB Documentation," Microsoft Azure, 2024. [Online]. Available: https://docs.microsoft.com/en-us/azure/cosmos-db/

[5] Microsoft Corporation, "Azure Key Vault Documentation," Microsoft Azure, 2024. [Online]. Available: https://docs.microsoft.com/en-us/azure/key-vault/

[6] Microsoft Corporation, "Azure Application Insights Documentation," Microsoft Azure, 2024. [Online]. Available: https://docs.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview

[7] HashiCorp, "Terraform Documentation," HashiCorp Developer, 2024. [Online]. Available: https://developer.hashicorp.com/terraform/docs

[8] Docker Inc., "Docker Documentation," Docker Docs, 2024. [Online]. Available: https://docs.docker.com/

[9] GitHub Inc., "GitHub Actions Documentation," GitHub Docs, 2024. [Online]. Available: https://docs.github.com/en/actions

[10] Node.js Foundation, "Node.js Documentation," Node.js, 2024. [Online]. Available: https://nodejs.org/en/docs

[11] R. Teal Witter and Lucy Li, "ResumeBoost: Evaluating the Effectiveness of AI Feedback on Resume Quality," arXiv preprint, 2023.

[12] National Association of Colleges and Employers (NACE), "Job Outlook 2024: Attributes Employers Want to See on New College Graduates' Resumes," NACE, 2024.

[13] Kubernetes Authors, "Kubernetes Documentation," Kubernetes.io, 2024. [Online]. Available: https://kubernetes.io/docs/home/

[14] Microsoft Corporation, "Azure Static Web Apps Documentation," Microsoft Azure, 2024. [Online]. Available: https://docs.microsoft.com/en-us/azure/static-web-apps/

APPENDICES

Appendix A – Pipeline YAML Files

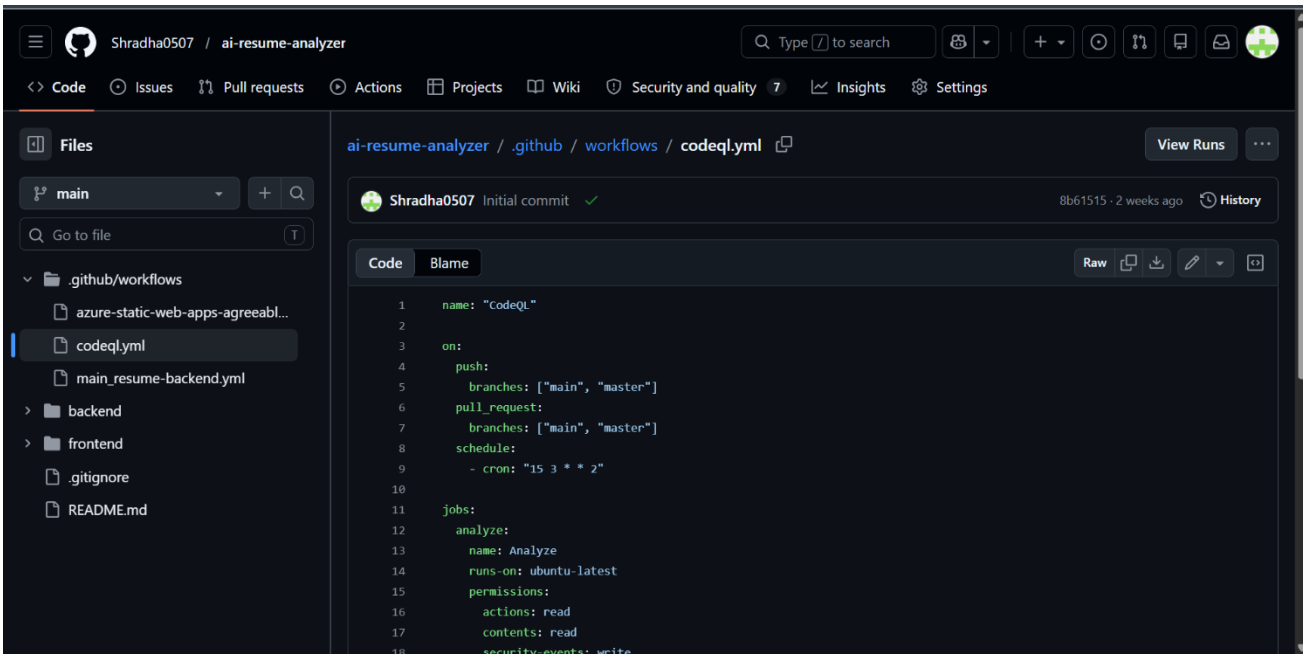

Figure A.1. Main Branch CI/CD Workflow YAML (ci-build.yml)

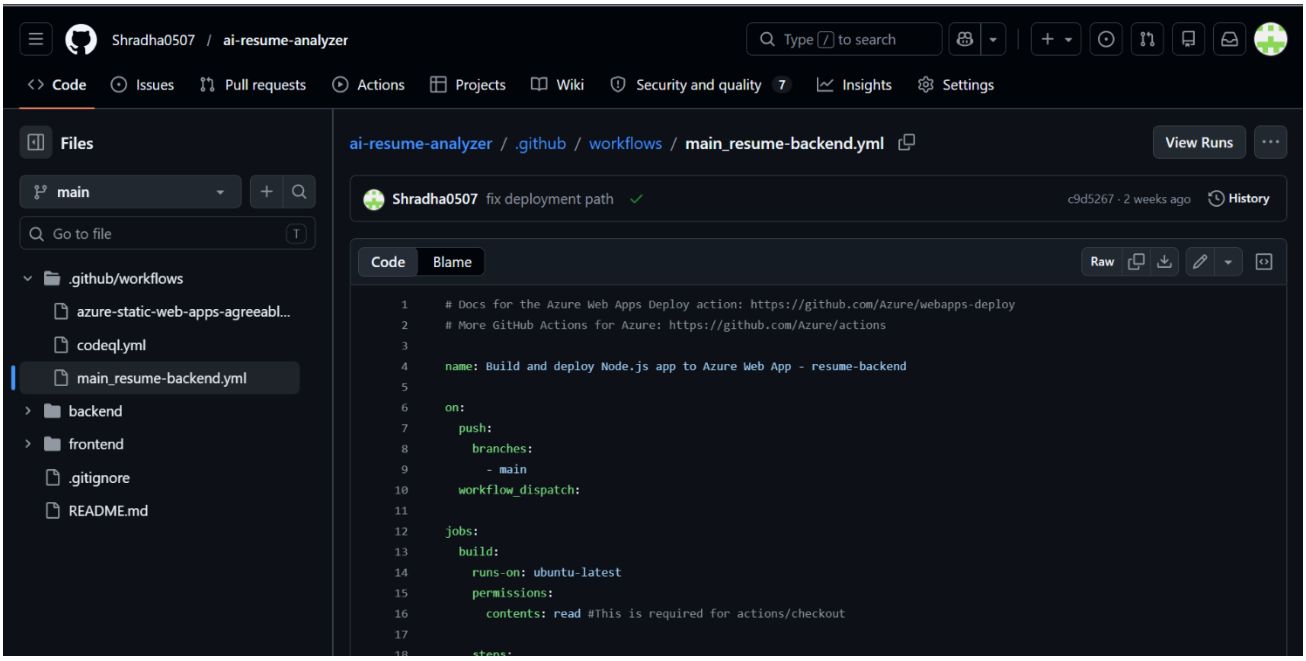

Figure A.2. Production Deployment Workflow YAML (prod.yml)

Appendix B – Screenshots (Deployments, AI Integration, Monitoring)

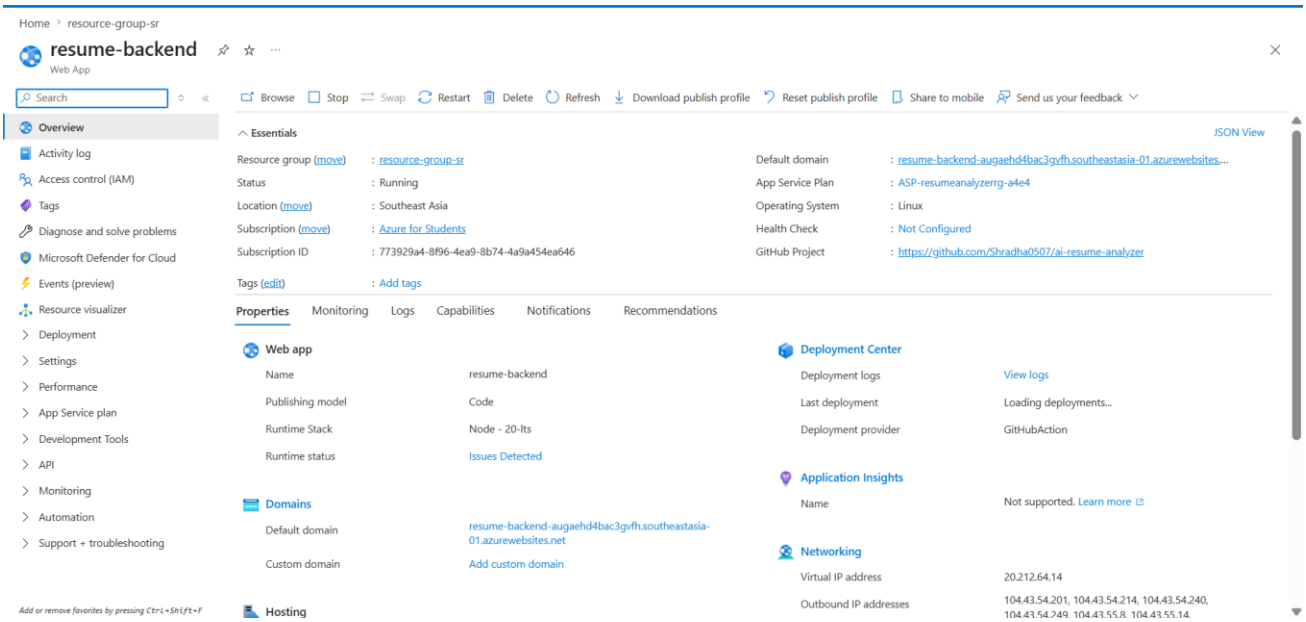

Figure B.1. Azure Portal – App Service Deployment Overview

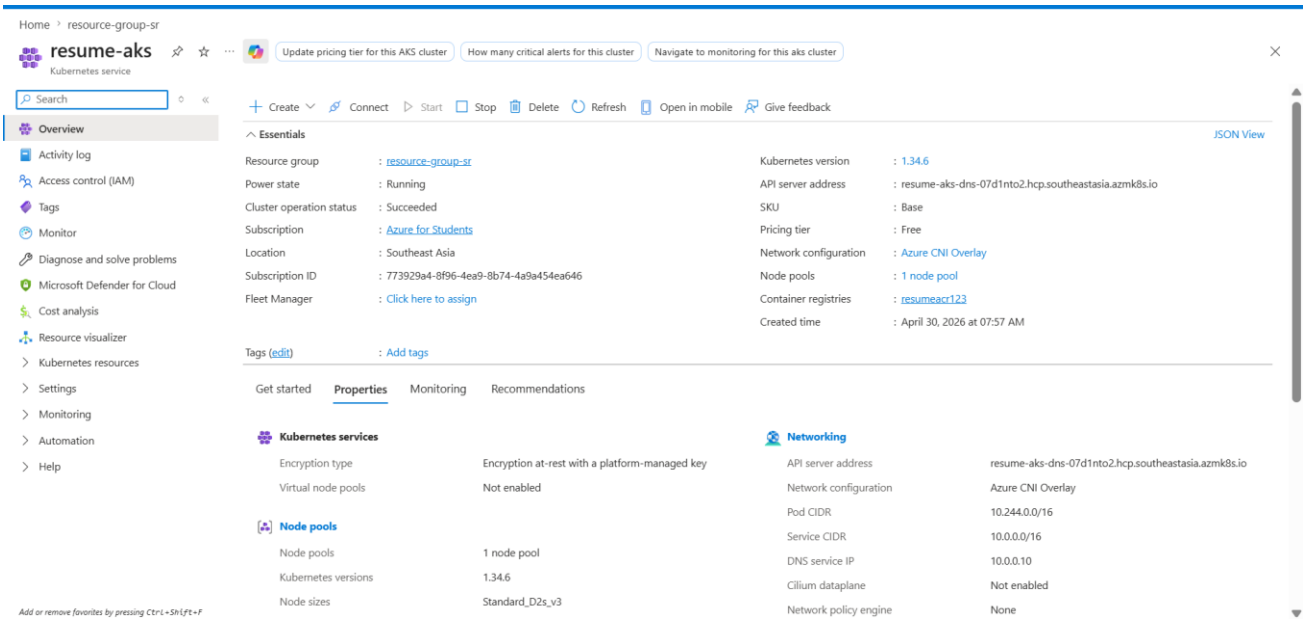

Figure B.2. Azure Portal – AKS

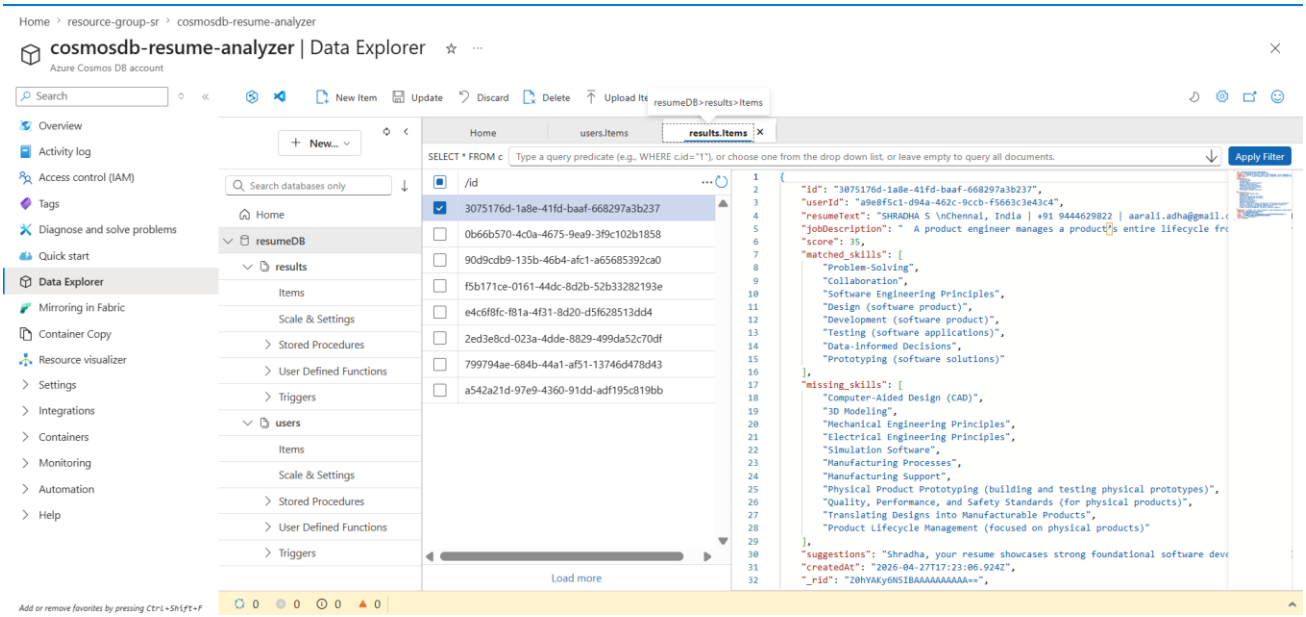

Figure B.3. Azure Portal – Cosmos DB Data Explorer

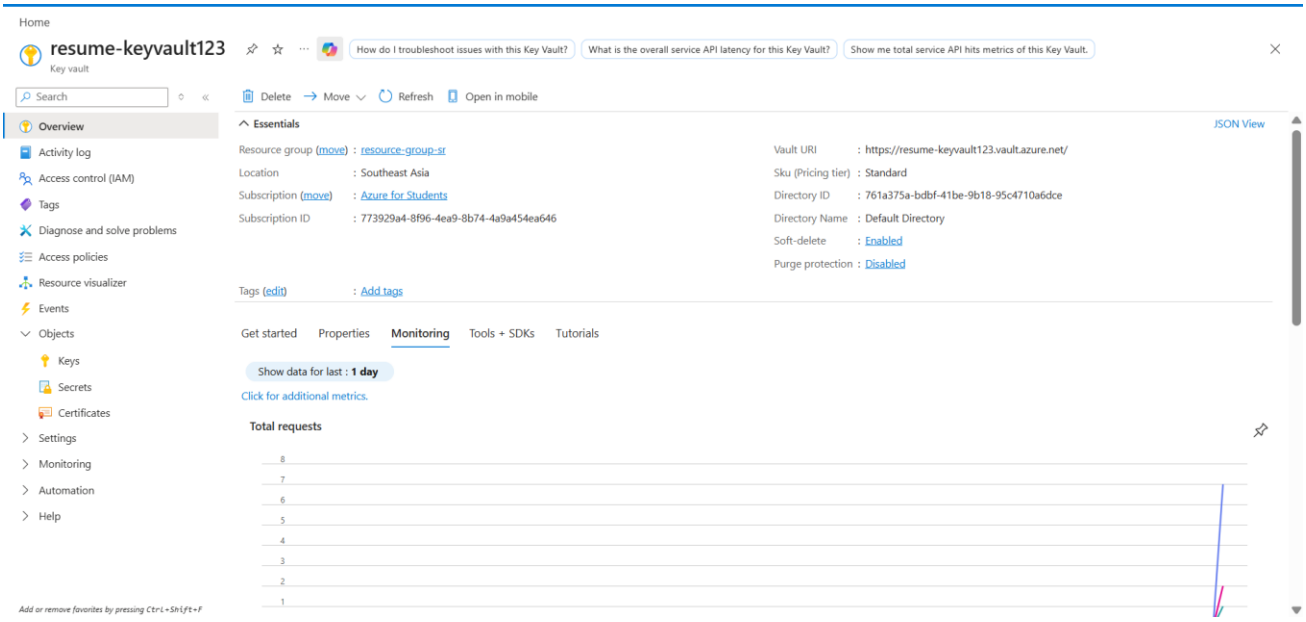

Figure B.4. Azure Portal – Key Vault Secrets List

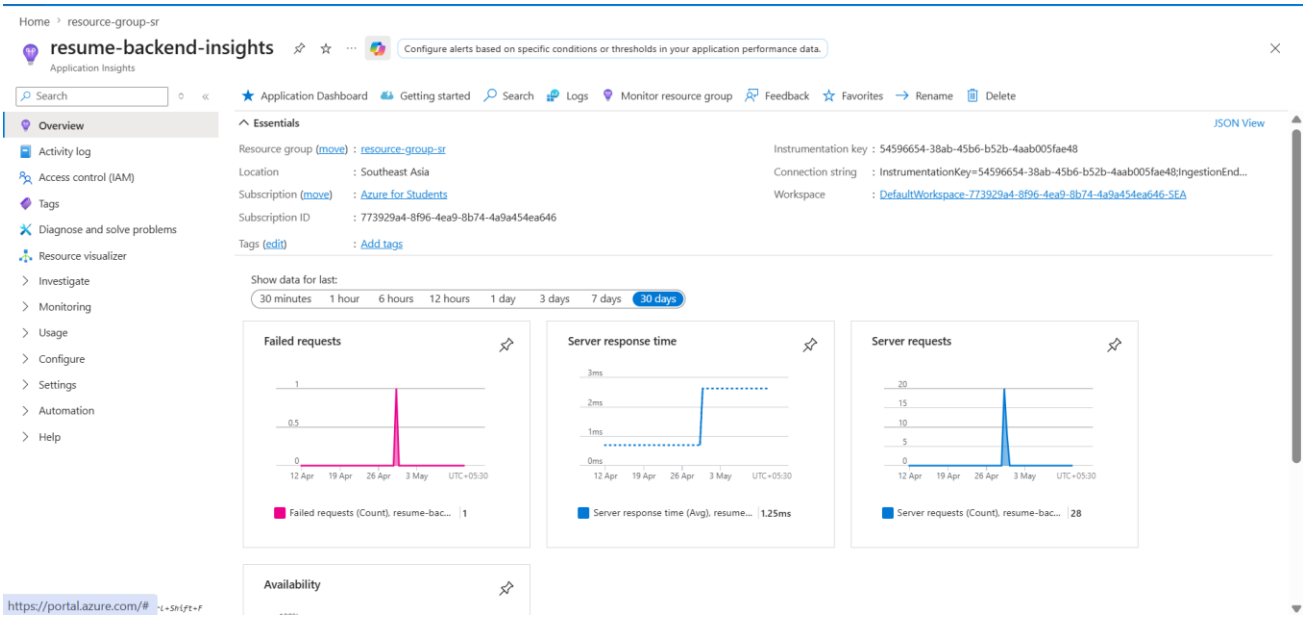

Figure B.5. Azure Application Insights – Live Metrics and Performance Dashboard

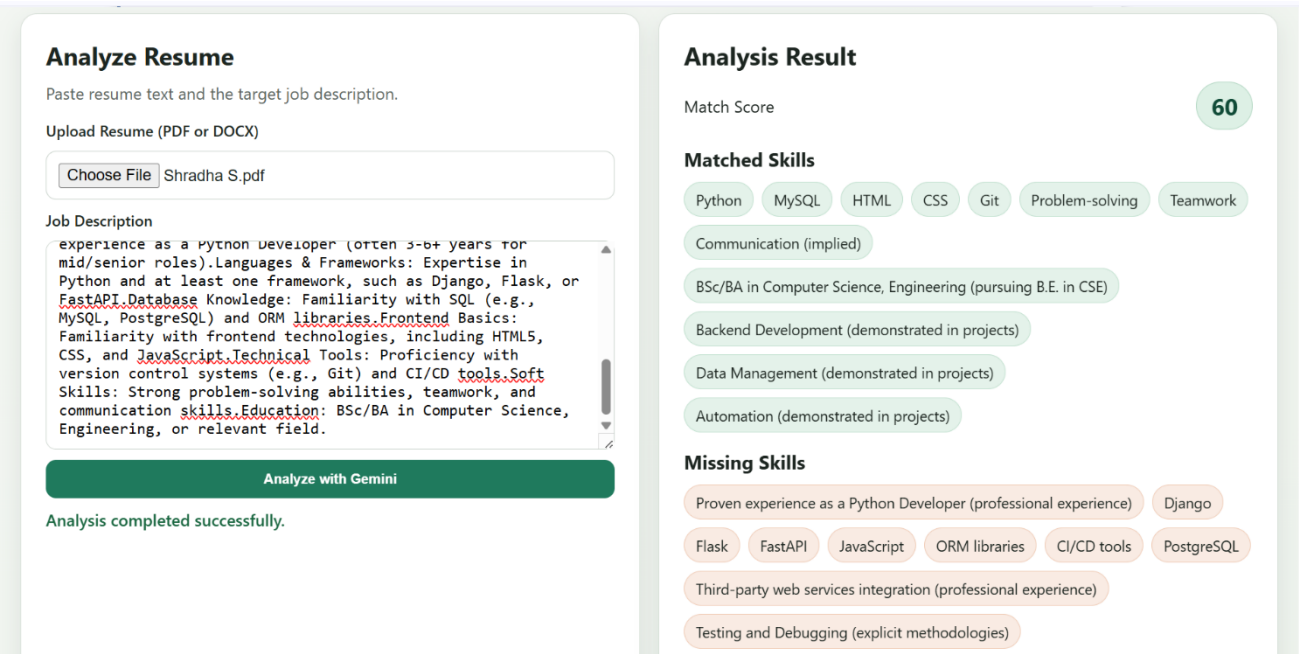

Figure B.6. AI Resume Analyzer – Analysis Results Display Page

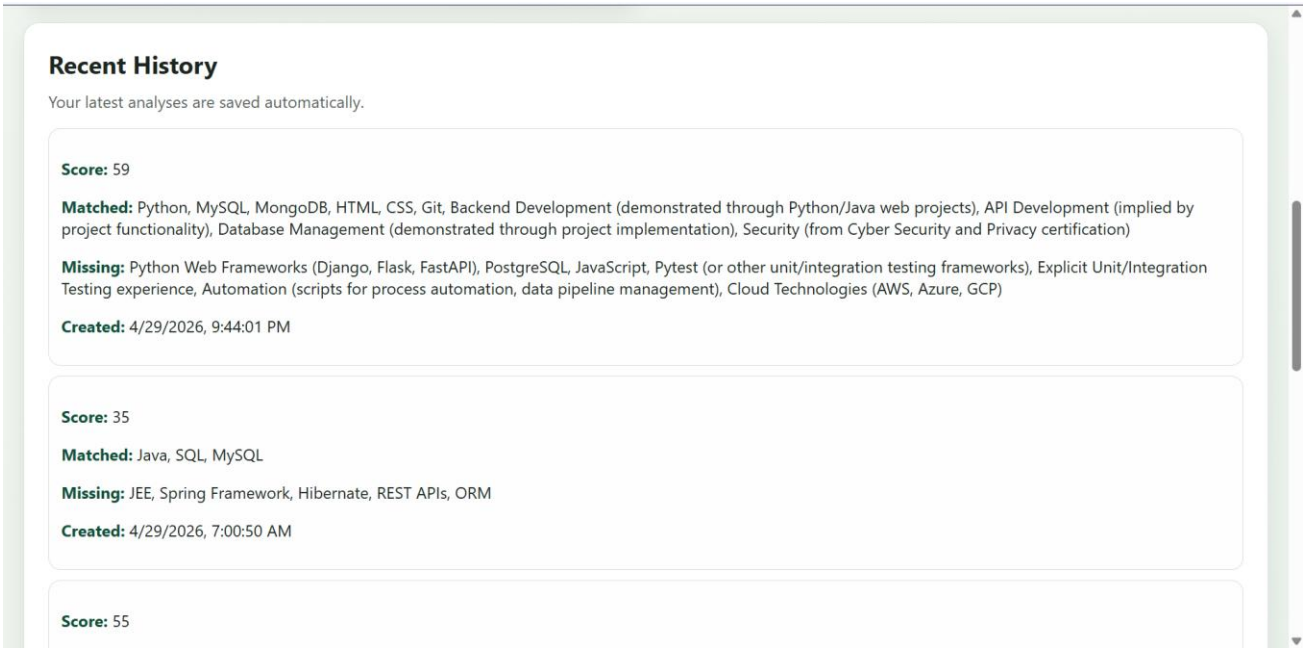

Figure B.7. AI Resume Analyzer – Analysis History Page

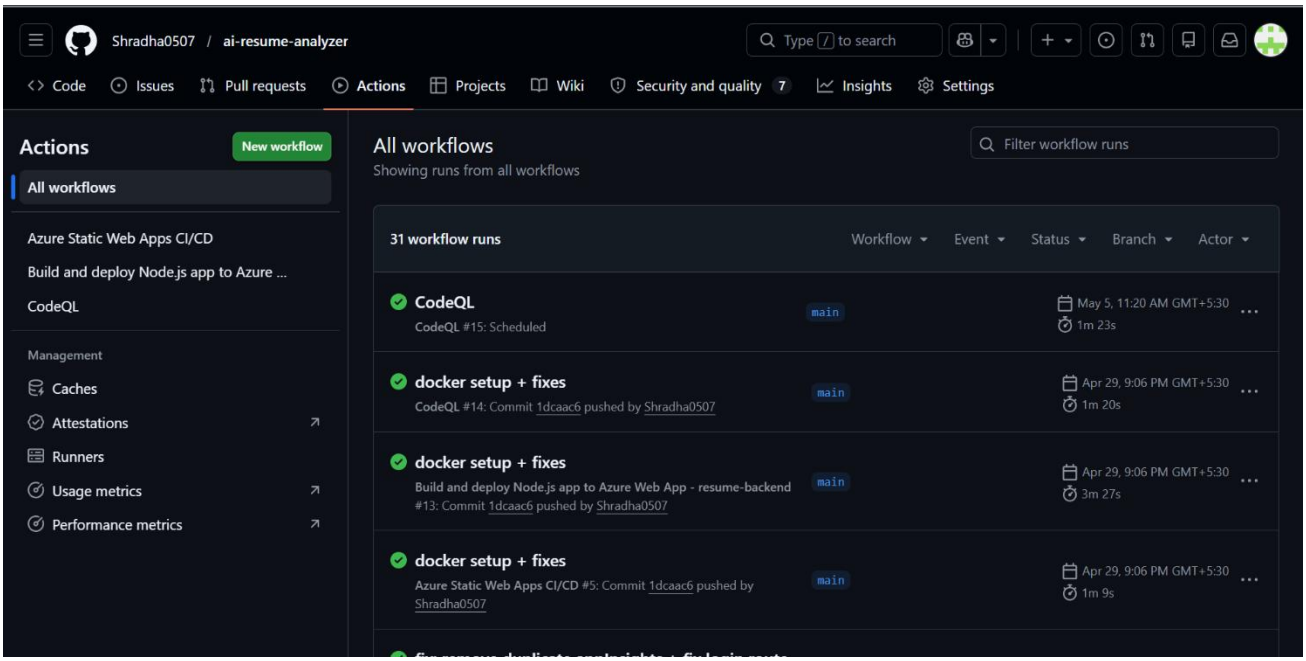

Figure B.8. GitHub Actions – Successful Full Pipeline Run